

THIS PCB's PURPOSE or DESIGN GOAL:

- * update MITS/Altair CPU-A board for modern EMC PCB layout principles
 - ** power plane(s) to reduce Vcc and Ground shifting niose
 - ** place decoupling capacitors next their respective ICs
- * improve processing speed above 1975 technology limited 8080A with newer CPU
 - ** HMOS 8085A upto 6MHz (late 1970s)
 - ** CMOS 80C85B upto 8MHz (mid 1990s)
- * add low voltage or 5Vcc < 4.7V forced RESET to help power up reliability
- * same PCB to support replacement of IMSAI 8080A MPU-A
 - ** have 2 ribbon cable connectors
 - ** S-100 jumpers and circuits for supporting signal variations
- * add Power On Jump to a user defined address (1K increments)
- * minimal hardware, no IO, no RAM, no ROM, etc... just CPU
- * allow for easy changing of CPU MHz speed
 - ** use DIP oscillator as flexible source
 - ** have PCB holes to allow HC case quartz crystal
- * design a spin off variation to use the NSC800 (a Z80A hybrid with 8085 like pin out)
- * No SMDs to keep a 1970s 1980s appearance

THIS DOCUMENT'S CONTENTS:

Change_History_and_Specs

File: S100_Card_sch.kicad_sch

S-100_Bus_Connector

File: S100_Bus.kicad_sch

Power_Supplies

File: Power_supplies.kicad_sch

8085CPU

File: CPU8085_to_S100.kicad_sch

Interrupt Vector Signals

File: Interrupts.kicad_sch

Power_On_Jump

File: POJMP_Port.kicad_sch

S-100_Address_and_Data_Bus (page 6)

File: AandDBus.kicad_sch

CPU_States

File: CPUstates.kicad_sch

ASSEMBLY INSTRUCTIONS

File: Assembly_sch.kicad_sch

CONFIGURATION OPTIONS

File: Options_sch.kicad_sch

VALIDATION PLANS

File: Validation_sch.kicad_sch

Proposed Configurations

File: Proposed_sch.kicad_sch

VALIDATION: RESET, SYS CLOCK AND PSYNC

File: Captures_sch.kicad_sch

VALIDATION: U24 CPU STATE DECODING

File: More_Captures_sch.kicad_sch

VALIDATION: THERMAL IMAGES

File: More_Captures3_sch.kicad_sch

VALIDATION: BUS DRIVER ENABLES

File: More_Captures2_sch.kicad_sch

Top Sheet, Repro update
Altair CPU-A with 8085A/80C85B

Sheet: /
File: CPU-85_r104.kicad_sch

Title:

Size: A4	Date: 11 August 2026, greglori_mi@wowway.com	Rev: rev 1.05
KiCad E.D.A. 10.0.4		Id: 1/17

CHANGE HISTORY:

6/13/2026 thru 7/4/2026 rev 1.0

- * started to make from IMSAI MPU-B rev 3.43 schematic by deleting parts
- * made new unique oscillator foot print to allow use of either 8 or 14 pin DIP package
- * changed all but 1 twin 2 pin jumpers (like J2 and J3) to 3 pin jumper
- * redesigned CPU state outputs to Front Panel based on MPU-A, CPU-A and MPU-B schematics
- * added 8 pin Altair Front Panel connector to support both IMSAI and MITS/Altair applications
- * added page 7 Power On Jump feature
- * 4 layer Gerbers made, no routing or DRC issues
- * no PCBs made

7/4/2026 rev 1.01

- * changed PCB to a 2 layer PCB
- * 2 layer Gerbers made, no routing or DRC issues
- * no PCBs made

7/8/2026 rev 1.02

- * added C26 for slower CPU crystals
- * 2 layer Gerbers made, no routing or DRC issues
- * changed 30u" (Military/Medical grade) to 20u" (Industrial grade) Hard Gold on S-100 fingers
- * schematic presented to S-100 Vintage Computer Group for review and comments
- * no PCBs made

7/9/2026 to 7/13/2026 rev 1.03

- * add CPU State Latch for driving CPU status sM1 on S-100 Bus DataOut5 for Front Panels
- * made new PSYNC signal, 3 selectable sources: IMSAI MPU-B ALE as PSYNC made a PSYNC 25% shorter; or M1 from CPU State Decoder; or a pulse lengthened to the end of T2 cycle
- * renumbered U23 thru U26, made space for added Tri State driver latch U23
- * two 8T97 changed to 74HCT367, worked OK on existing IMSAI MPU-B at 4MHz
- * newer KiCAD 10.99 was not compatible with FreeRouter 2.2.4, so had to re-load KiCAD 10.0.4
- * deleted D5 and R55 POJ indicator LED
- * FreeRouter 2.2.4 not performing autorouting operation (still investigating on 7/13/2026)
- * schematic given to S-100 Vintage Computer group for review and to identify issues
- * no PCBs made

7/14/2026 to 8/18/2026 rev 1.04

- * 8080 CPU State Table needs to be implemented not the 8085 CPU State Table in U23
- * U23 changed from 74HCT138 to 16V8 that can perform some complex gating
- * feedback sLNTA is now a hardwired 16V8 input because WINCUPL did not let me use an 16V8 internal path
- * restored D5 and R55 POJ indicator LED
- * auto routing problem with FreeRouter versions between 2.1.0, 2.2.0 and 2.2.4 stalls with unknown problem... Free Router team made aware of the version sensivity and given the DNS files to help debug this is FreeRouter Bug Report #753
- * success on 2 layer with Ground plane and 4 layer using older 2.1.0 Free Router
- * 2 and 4 layer versions PCB layouts made
- * only 2 layer Gerbers sent out for fabrication as 1.04.2 (18July2026 date in upper left corner)
- * andrasfuchs emailed issue #753 fixed in upcoming new FreeRouter release v2.3.x(?) (4Aug2026)
- * added new schematic page "Proposed Configurations" renumbered page 12 as 13

8/18/2026 to 10/6/2026 rev 1.05, correcting the ERRATA from 1.04.2 panel

- * useless "T2_LENGTH" old label name in silkscreen at old rev 1.01 J11 panel location, deleted this
- * renamed "2TLONGER" to "2xALE"
- * DIP switch labels are not on silk screen (but was on User Layer1), now moved back to top silk screen
- * Vector Interrupt "7" and S-100 finger label "10" too close, looks like "710", moved the "7" label
- * CPU Serial I/O label should be "SOD_SID" to position match pins, not "SID_SOD" (2 pin foot print is reversed)
- * IMSAI jumper label "FP" renamed to "NO FP" for showing that jumper is needed when there is not a Front Panel
- * re-arranged traces under the CPU for better and connecting the Ground planes under the CPU
- * corrected BOM, U10 is actually a 74HCT10, schematic page 7 was correct (U10 was 74HCT11 back at rev 1.02)
- * added test program to ASSEMBLY INSTRUCTIONS and a screen capture of a good test result
- * added page 14 for Measurements and Oscilloscope signal captures
- * added note on page 5 about X1 and X2 jumper use with 80C85B different than 80B5A
- * added page 15 for More Measurements and Oscilloscope signal captures
- * C31 too close to DIP Switch, DIP Switch is bigger than 16 pin DIP socket foot print, moved C31
- * moved /RESET trace away from R18 +5V solder pad for better clearance
- * consider connecting U1 /RESET to CPU output /POC, to avoid garbage from CPU (not known if this is an issue)
- * consider connecting U1 Q0 and Q1 instead of using CPU A0 and A1 to avoid garbage from CPU (not known if this is an issue)
- * U1 often resets to 1111 instead of 0000, could be the reset type, ordering 161 asynchronous reset version
- * changed silk screen labels for SWO signal to "SWO" and "/SWO" vs. X Y Z letter labels
- * based upon VR1 thermal image, more Ground plane exposed, 13mm right and left of VR1
- * consider adding jumpers for /POC /RESET and /EXT RESET, MITS ALTAIR, IMSAI and Cromemco are not all the same vs IEEE-696
- * U19 changed to PLD16V8 to better control and prevent AD Bus contention
- * U11 pin 4 connected to /ALE to avoid AD Bus contention when lower address is time multiplexed
- * most resistors combined into resistor packs to reduce the number of sold joints
- * J5 is connected to pin 72 XRDY, not 5, schematic page 8 is correct, text on page 11 corrected

Reference Specifications:

Mechanical Outline is from IEEE 696 S-100 Bus Specification 1983-06_Images.pdf

NOTE1: Bottom S-100 Edge, 45 degree chamfer, 0.60mm into the top and bottom faces, 1/3 depth or 0.6mm of PCB thickness; total PCB thickness 1.60mm to 1.80mm or 0.066inch

NOTE2: 20u" Hard Gold on S-100 fingers with "lmm" finish

NOTE3: PCB material is FR4 - 170TG with >200V isolation, UL 94V-0 Flammability

NOTE4: 2oz copper front and back layers, inner layers 1oz
Green LPI masking, via holes plugged with masking
white silk screen text

NOTE5: IMSAI MPU-B Schematic Rev 2.pdf as updated by Robret Weatherford 3/2021

NOTE6: IMSAI MPU-B Reference Manual

NOTE7: page 24 880-101 CPU-A card schematic from MITS_Altair8800TheoryOperation_1975.pdf

NOTE8: page 34 of S-100 Bus Handbook by Dave Bursky, Hyaden Book Company, Rochelle Park, NJ

NOTE9: <https://www.televiretech.com/blogs/technical-resources/the-definitive-guide-to-connector-plating-thickness-15%CE%BC-gold-30%CE%BC-gold-and-tin-compared>

NOTE10: Cromemco ZUP (Z80A) card schematic

NOTE11: IMSAI Schematics, Page 4 is MPU-A rev4 from 2/76

NOTE12: Intel MCS80 8080 Microcontroller Systems User's Manual September 1975

SOFTWARE VERSIONS USED:

KiCad

Version: 10.0.4, release build

wxWidgets 3.3.2 Unicode and Boost 1.90.0
Platform: Windows 10 (build 19045), 64-bit edition, 64 bit
GDI objects in use 696

Freerouting - an open-source PCB auto-router

Version: 2.1.0

This program comes with no warranty!

Latest release: <https://github.com/freerouting/freerouting/releases>

Report a problem: <https://github.com/freerouting/freerouting/issues>



S-100 PCB Card Information

Sheet: /Change_History_and_Specs/

File: S100_Card_sch.kicad_sch

Title:

Size: A4

Date: 6 October 2026, gregori_mi@wowway.com

Rev: rev 1.05

KiCad E.D.A. 10.0.4

Id: 2/17

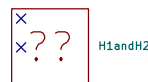
J1
S100_MALE_as_2_columns

1	1	+8_V	+8_V	51	51
2	2	+16_V	-16_V	52	52
3	3	XRDY	SSWDSB*	53	53
4	4	VIO*	EXT_CLR*	54	54
5	5	VI1*	TMA0*	55	55
6	6	VI2*	TMA1*	56	56
7	7	VI3*	TMA2*	57	57
8	8	VI4*	sXTRQ*	58	58
9	9	VI5*	A19	59	59
10	10	VI6*	SIXTN*	60	60
11	11	VI7*	A20	61	61
12	12	NMI*	A21	62	62
13	13	PWRFAIL*	A22	63	63
14	14	TMA3*	A23	64	64
15	15	A18	NODEF2	65	65
16	16	A16	NODEF3	66	66
17	17	A17	PHANTOM*	67	67
18	18	SDSB*	MWRT	68	68
19	19	CDSB*	PROT_STAT*	69	69
20	20	UNPROT	PROT	70	70
21	21	DIDSBL*	RUN	71	71
22	22	ADSB*	RDY	72	72
23	23	DODSB*	INT*	73	73
24	24	Phase2	pHOLD*	74	74
25	25	Phase1	RESET*	75	75
26	26	pHDLA	pSYNC	76	76
27	27	pWAIT	pWR*	77	77
28	28	INTE	pDBIN	78	78
29	29	A5	A0	79	79
30	30	A4	A1	80	80
31	31	A3	A2	81	81
32	32	A15	A6	82	82
33	33	A12	A7	83	83
34	34	A9	A8	84	84
35	35	DO1_BD1	A13	85	85
36	36	DO0_BD0	A14	86	86
37	37	A10	A11	87	87
38	38	DO4_BD4	DO2_BD2	88	88
39	39	DO5_BD5	DO3_BD3	89	89
40	40	DO6_BD6	DO7_BD7	90	90
41	41	DI2_BD10	DI4_BD12	91	91
42	42	DI3_BD11	DI5_BD13	92	92
43	43	DI7_BD15	DI6_BD14	93	93
44	44	sM1	DI1_BD9	94	94
45	45	sOUT	DIO_BD8	95	95
46	46	sINP	sINTA	96	96
47	47	sMEMR	sWO*	97	97
48	48	sHLTA	ERROR*	98	98
49	49	CLOCK	POC*	99	99
50	50	OV_GND	OV_GND	100	100

NOTE: S-100 Labels and PCB Holes Cut & Pasted from John Monohan's web page https://avitech.com.au/?page_id=3240 (Google Vintage Computer Group, used on IMSAI CPU-A repro panel) 26April2025

That symbol re-imported into KiCad, and the symbol edited from 1 column of 100 to 2 columns of 50 format to allow fitting on A sized page; DI, DO and 16 bit pins renamed per page 18 of IEEE-696 standard; a few pin names corrected ei. IEEE called pins 92 and 93 BiDirectional 3 "even" and BiDirectional 4 "odd"... which looks to be typo errors to me. 22June2026

Card ejector mounting holes



NOTE: S-100 pin number labels are used to connect to circuits on these schematic pages.

S-100 Connector Assignments

Sheet: /S-100_Bus_Connector/
File: S100_Bus.kicad_sch

Title:

Size: A4

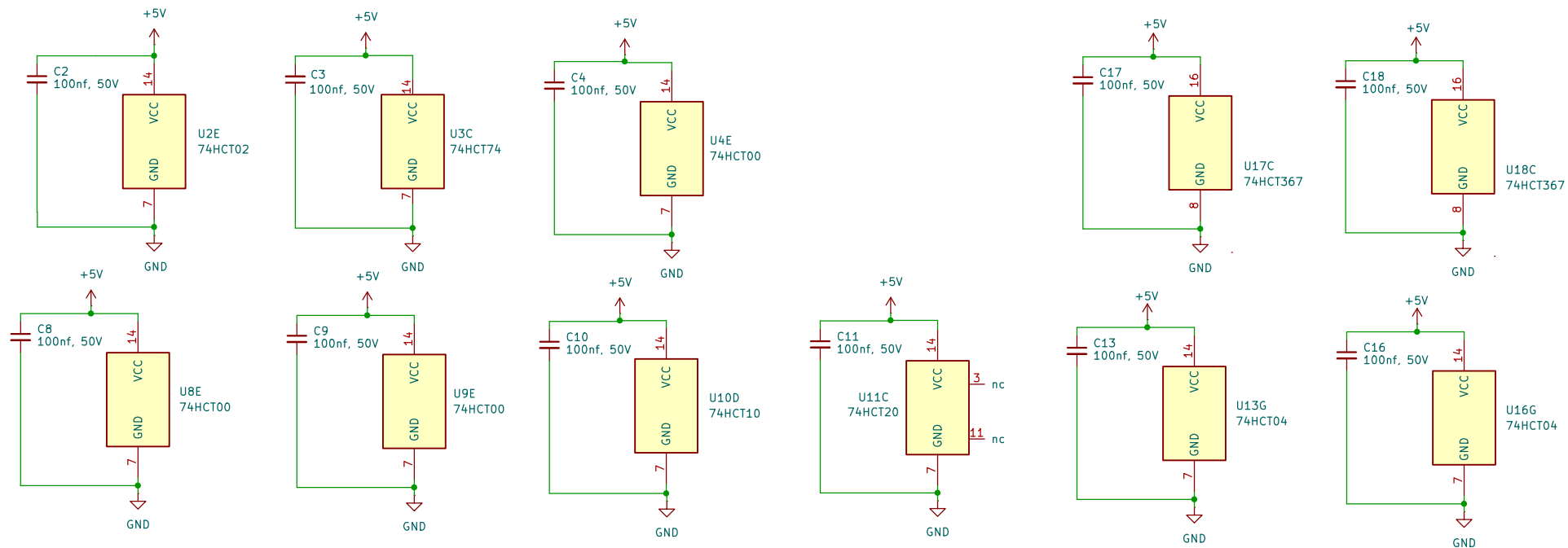
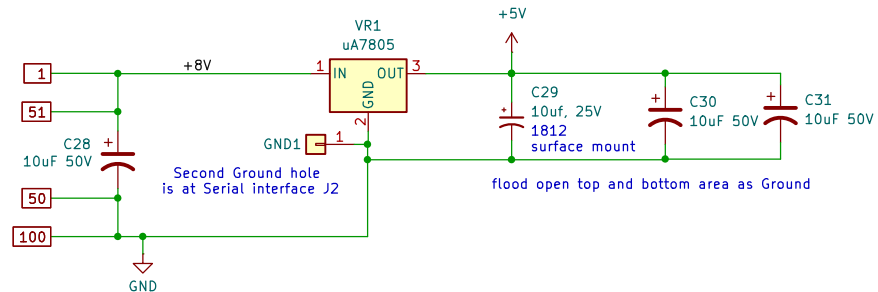
Date: 17 July 2026, greglori_mi@wowway.com

Rev: rev 1.05

KiCad E.D.A. 10.0.4

Id: 3/17

Note: Use 4 or more vias to connect PCB Top to PCB Bottom copper on +8V and Ground



locate each decoupling capacitor at each respective IC location Vcc pin position

5V Regulator & Decoupling Caps

Sheet: /Power_Supplies/
File: Power_supplies.kicad_sch

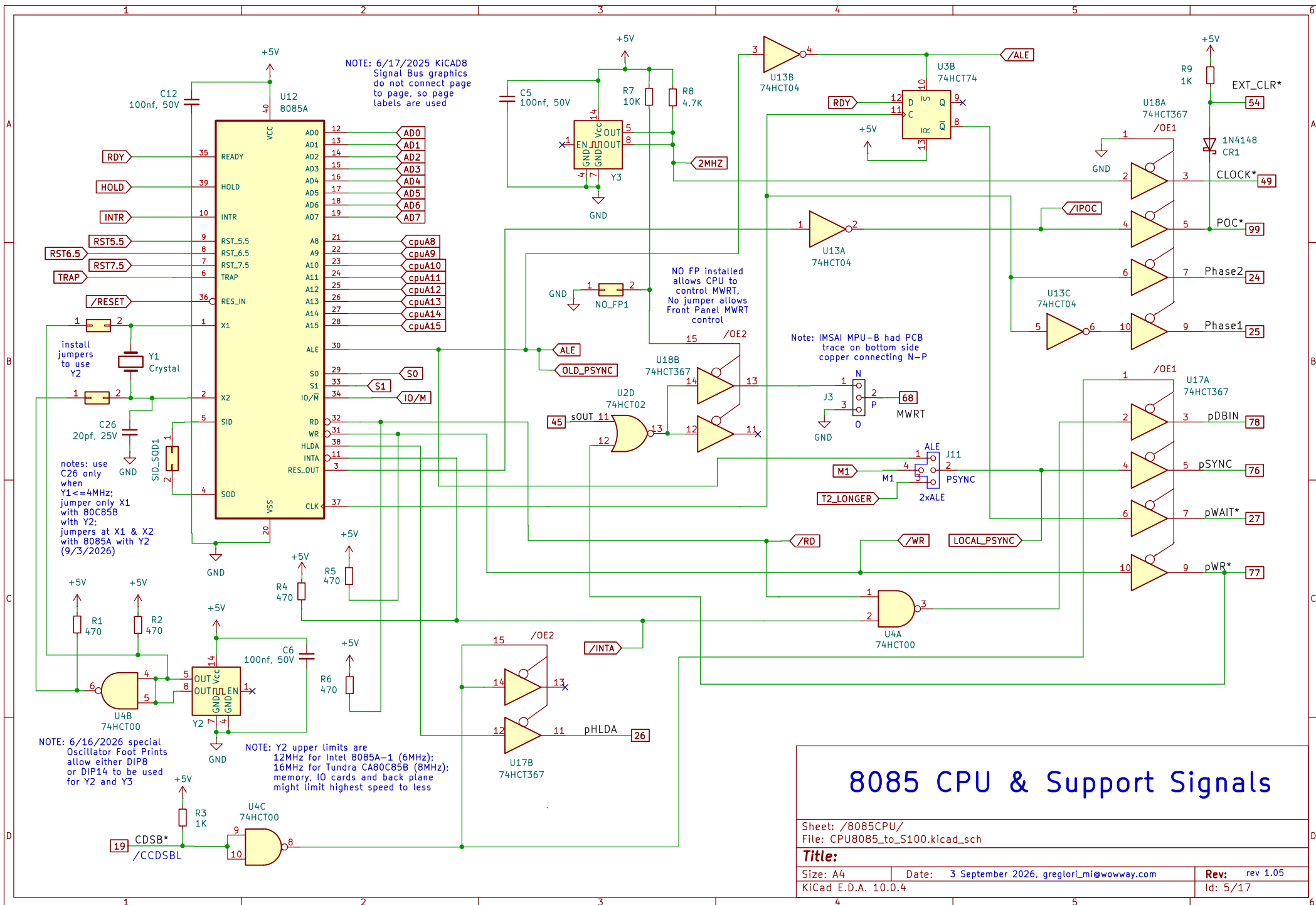
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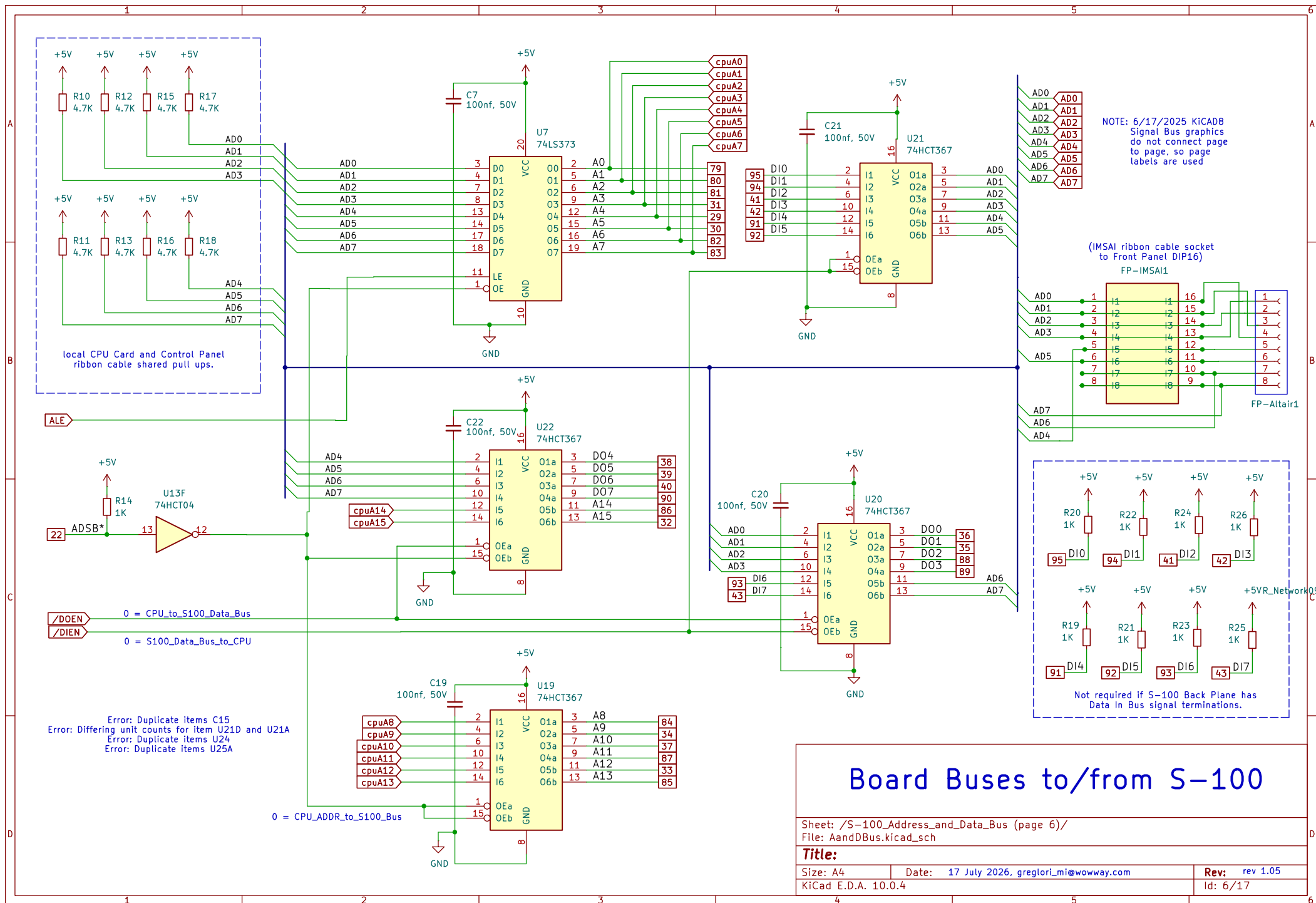
Size: A4 Date: 17 July 2026. greglori_mi@wowway.com

KiCad E.D.A. 10.0.4

Rev: rev 1.05

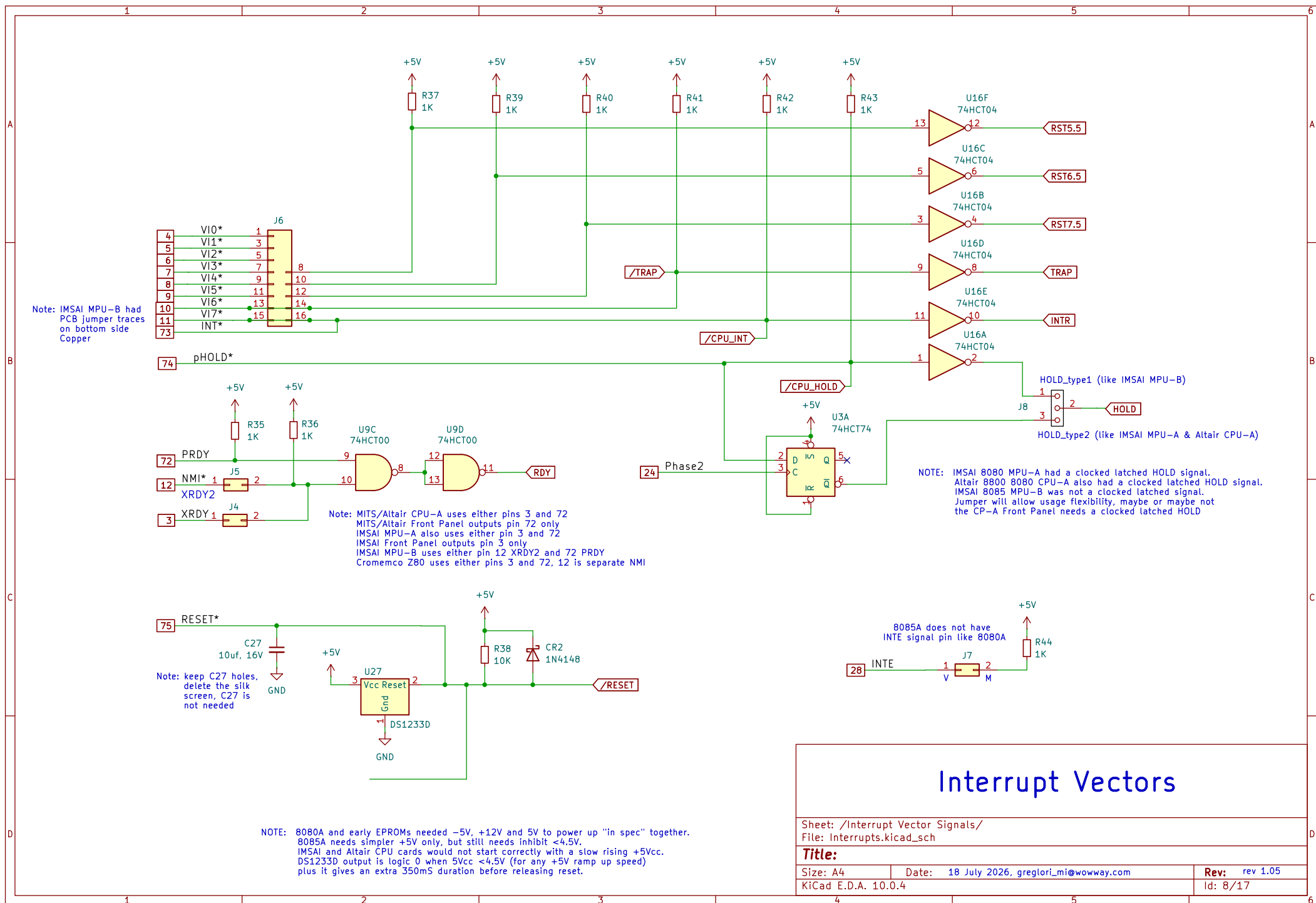
Id: 4/17

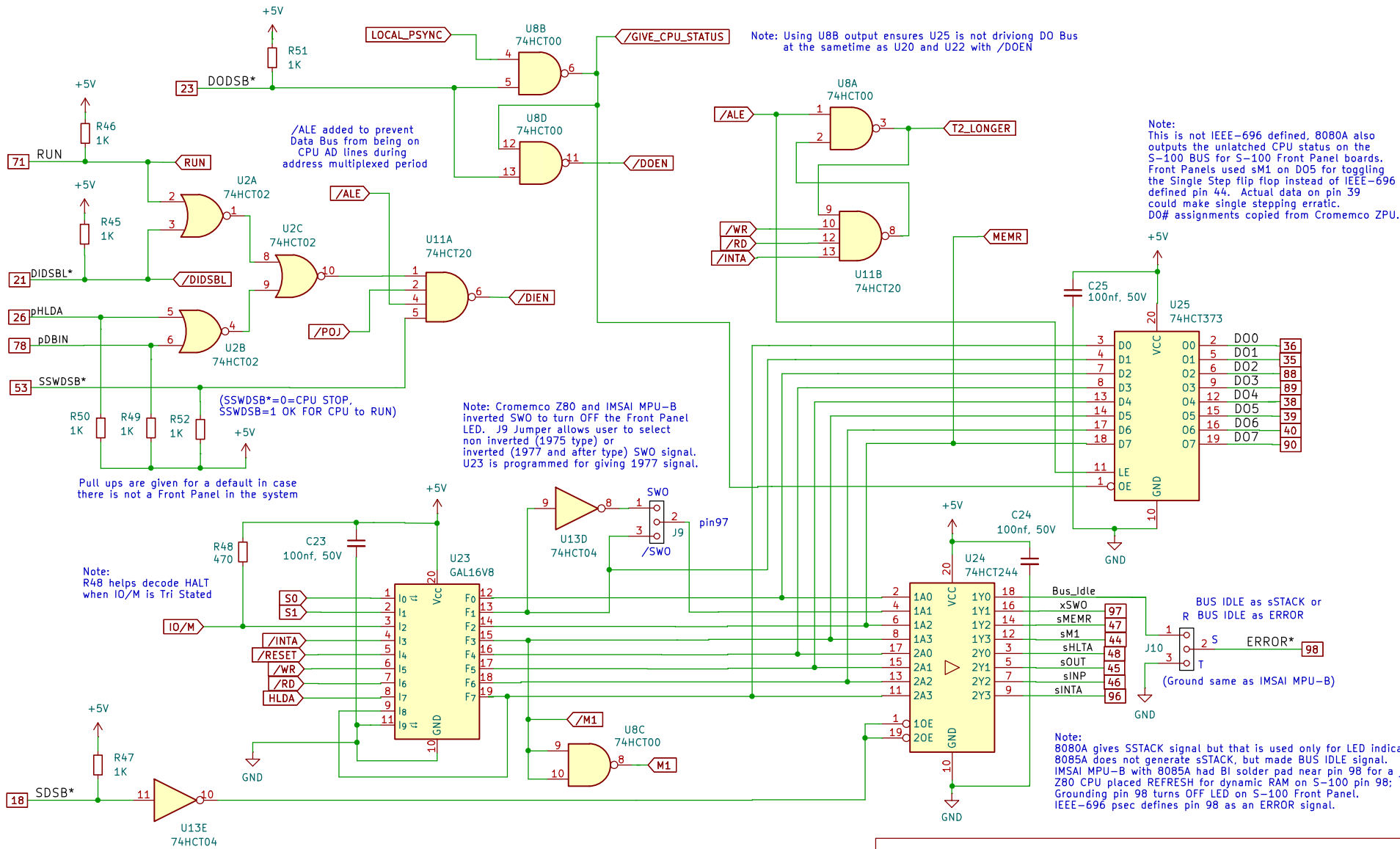




[illegible]

Id: 7/17





CPU State Decoding

Sheet: /CPU_States/
File: CPUstates.kicad_sch

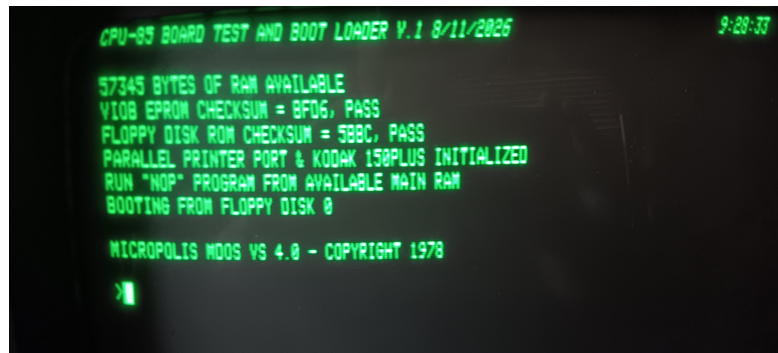
Title:

Size: A4 Date: 4 October 2026, greglori_mi@wowway.com
KiCad E.D.A. 10.0.4

Rev: rev 1.05
Id: 9/17

ASSEMBLY INSTRUCTIONS:

- 1) solder lowest height parts first... resistors, axial diodes
- 2) then solder DIP sockets, the ceramic capacitors, LED
- 3) and then the 2 and 3 pin SIP headers
- 4) solder tallest parts radial electrolytic capacitors and the VR1 & its optional heat sink last
- 5) Inspect all solder joints with a 5x to 10x magnifying glass.
- 6) use black electrical tape to cover gold electrical edge card fingers and any front to back PCB holes to prevent PCB cleaner seeping under front side parts
- 7) hold PCB vertically and apply about 3 oz, or 1 even coating, over the rear of the PCB of the ZEP Citrus cleaner
- 8) wait about 1 minute
- 9) hold PCB vertically, apply about 3 oz to 4 oz or 1 even washing coat of Kwik Clean Flux Cleaner
- 10) wait about 1 minute
- 11) rinse with distilled water
- 12) shake water off the PCB or use a hair dryer on "Cold" high speed setting to blow dry the PCB
- 13) power up the panel without ICs to check the regulator output voltage (5V)
- 14) without power applied to the panel, install ICs logic, CPU, etc. (Do not "Hot Plug" parts or do not install parts with power active.)
- 15) looking down onto the top of the PCB, look for any outward bent pins that did not go into their socket receivers, remove, fix, un-bend and reinstall parts if necessary
- 16) look from the PCB S-100 edge under all ICs, check for any unseen bent, folded under pins, remove, fix, un-bend and reinstall parts if necessary
- 17) put a drop of rubber cement (not permanent, do not use epoxy) under DIP switch and DIP oscillators to help hold them in place
- 18) set DIP switches and jumper blocks to desired configuration
- 19) power up the CPU-85 and go.... use as needed with your S-100 application(s)
- 20) make a test program to exercise Printer, ROM and RAM memory, Video and Floppy disk. Run this program to gain operational confidence in your specific system.
My test program can be found at <https://github.com/Puppy-the-CoPilot>
This particular test program does the following:
 - * runs from EPROM at E000H
 - * tests RAM from 0000H to DFFFH (57K)
 - * will run a test program of 57K in series No Operation instructions
 - * displays the testing status at F000H Video RAM
 - * checks Video Terminal Emulation ROM at F800H and Floppy ROM at EC00H
 - * displays Real Time Clock Value HH:MM:SS on the video display
 - * initializes an ink jet printer, and requests printing of 1 text line
 - * boot from floppy disk 0 (complex move head to track 0, read the disk, and execute "RES"ident OS)
 - * "PASS" if all above checks PASS, and video monitor displays "MICROPOLIS MDOS VS 4.0 - COPYRIGHT 1978" and entry prompt ">"



ASSEMBLY INSTRUCTIONS

Sheet: /ASSEMBLY INSTRUCTIONS/
File: Assembly_sch.kicad_sch

Title:

Size: A4	Date: 18 August 2026, gregori_mi@wowway.com	Rev: rev 1.05
KiCad E.D.A. 10.0.4	Id: 10/17	

CONFIGURATION OPTIONS:

J11 PSYNC source
* use ALE address latch like IMSAI MPU-B
* use M1 from CPU State Decoder (like Cromemco Z80 board), warning M1 can be valid during immediate operand data byte collection till end of T6
* use 2x ALE long signal lasting till end of T2 cycle like a 8080A on IMSAI and MITS/Altair boards (recommended)

J8 HOLD circuit source used for systems with multiple CPUs and/or DMA functions
* Type1, no latch like IMSAI MPU-B
* Type2, latched like MITS/Altair CPU-A
see Intel 8085 data sheet page 6 Fig. 6

XRDY and XRDY2 sources
* J4 jumper uses pins 3 and 72 like MITS/Altair, IMSAI MPU-A and Cromemco Z80 (recommended)
* J5 jumper uses pins 12 and 72 like IMSAI MPU-B (unusual configuration)

J6 Interrupts 5.5, 6.5 and 7.5
note: 8085 contains 3 vectors of a 8214
* jumpers select 3 interrupt sources 1 thru 8

FP1, drive source for MWRT S-100 bus signal pin 68
* No jumper lets the Front Panel drive MWRT
* If no Front Panel, then install this jumper to let CPU drive MWRT this is MWRT Enable (J3 on IMSAI MPU-B)

J3 MWRT Source S-100 pin 68
* CPU driven signal (N-P jumper) also install FP1 for MWRT Enable
* Ground (P-O jumper)

J7 INTE S-100 pin 28 (V-M jumper)
note: 8085 does not have an INTE output
* Open Circuit or
* pull up voltage to +5V (like IMSAI MPU-B)

install Y2 = maximum allowed by users system when Y1 is not installed

Solder pads for SID and SOD of 8085

DS1233 +5V Monitor, Low Voltage Inhibit (LVI)
note: almost all CPUs from 1970s lack LVI, MicroChip and Motorola made LVI a standard feature in late 1980s
* do not populate if 1970s performance is desired (not recommended)
(recommend installing to avoid start up and shut down glitching issues on floppy drives)
* no C31 makes a squared off, fast rising edge; >300mS delay from DS1233D exceeds Xtal oscillator start up time

Y1 / Y2 Jumpers
* no jumpers, allows use of Y1 quartz crystal per Intel 8085 data sheet page 6 figure 5
* or DIP oscillator Y2 as CPU clock source (my preference is for Y2 DIP oscillator)

Ribbon Cable Interfaces
* 16 pin PDIP for IMSAI Front Panel
* 8 pin Molex for MITS/Altair Front Panel

Power On Jump, DIP switches
* switch 1 = ON enables Power On Jump at Reset
* switch 1 = OFF uses CPU default 0000H address
* switch 2, not used
* switch 3 = Address A10 to switch 8 = Address A15, ON=logic 0, OFF=logic 1 (1K increments from 0400H to FC00H)

J9 SWO signal source for S-100 pin 97
* IMSAI MPU-B and Cromemco Inverted xSWO to turn OFF Front Panel LED as default for 1977 and after (a 1977 corrective action)
* IMSAI MPU-A MITS/Altair type xSWO signal (1975 jumper) as implemented in 1975, but will leave Front Panel LED ON by default (an IMSAI MITS/Altair oops???)
This signal might not have been used by any S-100 board other than the Front Panel LED. (aka. this might be a don't care)

ERROR or SSTACK source for S-100 pin 98
note: 8085 does not make the SSTACK output
* Ground to disable Front Panel LED (S-T jumper) like IMSAI MPU-B
* or decoded BUS IDLE CPU State (R-S jumper)
* MITS/Altair CPU-A and IMSAI MPU-A called this SSTACK
* IEEE-696 defines 98 as bus ERROR signal
This signal too might be a don't care and not used by any S-100 board.

PCB OPTIONS

Sheet: /CONFIGURATION OPTIONS/
File: Options_sch.kicad_sch

Title:

Size: A4 Date: 6 October 2026, greglori_mi@wowway.com

KiCad E.D.A. 10.0.4

Rev: rev 1.05

Id: 11/17

PRELIMINARY: THIS PAGE NEEDS CHECKING IN ACTUAL SYSTEMS. RECOMMENDED JUMPER CONFIGURATIONS:

No Front Panel S-100 Systems (like my IMSAI PCS80/30):

- 1) Since my PCS 80/30 lacks a Front Panel to enter a PC value to run from, I wrote a test program into an EPROM starting at E000H.
(note: I knew this when I started so this an additional reason why I added the Power On Jump feature to CPU-85.)
- 2) Set Power On Jump to go to users Boot ROM (DIP Switches set to ENABLE POJ=ON and EOH, ECH if I want the Micropolic Floopy Boot program)
- 3) Y1 and C26 not installed, jumpers installed to use Y2=2MHz oscillator until I determine highest feasible MHz for my system.
- 4) NO_FP jumper installed to allow CPU to control MWRT, MWRT jumper N-P; else install this jump to allow Front Control Panel to toggle MWRT
- 5) J4 jumper installed, leave J5 open or no jumper block.
- 6) J6 no interrupts or jumpers used (user's application specific).
- 7) J7 jumper to have pull up to 5V for INTE (same as IMSAI MPU-B).
- 8) J8 jumper HOLD_TYPE1 closed (same as the IMSAI MPU-B).
- 9) J9 jumper X-Y for not inverting (U23 output is programmed to make the inverted /sW0 for 1977).
- 10) J10 jumpered S-T to Ground (original IMSAI MPU-B setting).
- 11) J11 use 2T_LONGER signal as PSYNC source.

Front Panel S-100 like IMSAI 8080A and MITS/ALTAIR 8800:

- 1) Set Power On Jump-OFF (unless you want to go to a Boot program)
- 2) Y1 and C26 not installed, jumpers installed to use Y2=2MHz oscillator until you determine highest feasible MHz for your system.
- 3) No_FP jumper block installed to allow the Front Panel to control MWRT
- 4) J4 jumper block only, for 1975 pre IEEE696; J5 open or no jumper block
- 5) J6 no interrupts or jumpers used (user's application specific)
- 6) J7 jumper to have pull up to 5V for INTE.
- 7) J8 jumper HOLD_TYPE2 closed (same as the IMSAI MPU-A and MITS/ALTAIR CPU-A).
- 8) J9 jumper X-Y for having /sW0 the same state as what it was in 1975 (which turned the Front Panel LED ON when it should have been OFF).
- 9) J10 jumpered R-S (sSTACK is like Bus Idle signal, this just drives the Front Panel LED).
- 10) J11 use 2T_LONGER signal as PSYNC source to make it as long as the 8080A signal.
- 11) Connect the Front Panel cable to/from the CPU-85 card

With Don Caprio IMSAI 8080 3/28/2022 CP-A Repro Front Panel:

To Use the reproduction Front Panel on my PCS 80/30 I needed to use an extension card to lift the Front Panel card above the Chassis frame.
web page <http://s100chassis.pbworks.com/w/page/148514070/FrontPage> deosn't recommend all DIP switch positions.

- 1) Front Panel DIP Switch #1=OFF, UNPROT
- 2) Front Panel DIP Switch #2=ON, UNPROT GND (pin 20 is cut, so this should not matter)
- 3) Front Panel DIP Switch #3=OFF, MEMPROT (don't protect RAM)
- 4) Front Panel DIP Switch #4=OFF, TERM PSYNC
- 5) Front Panel DIP Switch #5=ON, PSYNC
- 6) Front Panel DIP Switch #6=OFF, sM1
- 7) Front Panel DIP Switch #7=ON, 8080
- 8) Front Panel DIP Switch #8=OFF, Z80
- 9) Front Panel DIP Switch #9=ON, MWRT (let the Front Panel generate MWRT)
- 10) Front Panel DIP Switch #10=OFF, SSW_DSBL (dont disable, OK send data via the ribbon cable???)
- 11) S-100 pin #20 cut disconnected per email "IEEE CP-A Compatibility Issue" dated 9/3/2025
- 12) S-100 pin #53 cut snd un-Grounded per email "IEEE CP-A Compatibility Issue" dated 9/3/2025
- 13) S-100 pin #70, keep it Grounded, no cut to trace at this pin.
- 14) 10uF increased to 20uF, 1000pf increased to 10nF, both next to U17 on the Front Panel board
- 15) proceed with "No Front Panel S-100 Systems (like my IMSAI PCS80/30)" above

With Jade Bus Probe:

- I used the Jade board for debugging and validating U23 CPU state decoder.
- 1) Setings from "No Front Panel S-100 Systems" above
 - 2) All jumpers Open or No Installed (UNPROT and PROT also open) on the Jade Bus Probe.

PROPOSED CONFIGURATIONS

Sheet: /Proposed Configurations/

File: Proposed_sch.kicad_sch

Title:

Size: A4

Date: 6 October 2026, greglori_mi@wowway.com

Rev: rev 1.05

KiCad E.D.A. 10.0.4

Id: 12/17

VALIDATION SET UP:

(Validation Set Ups can be critical)

- 1) PCS 80/30 Card Cage: (In case card position is important)
 - Slot#1 = VIO-B Video card (350nS, expect higher speed problems)
 - Slot#2 = Tanner 64K RAM (150nS), 56K used, upper 8K of RAM disabled for VIO-B and Micropolis cards
 - Slot#3 = open (sometimes the extender card, sometimes a PROM100 Plus EPROM Programmer)
 - Slot#4 = 32K of Boot ROM, extended memory card #1 (has Micropolis MDOS), disabled for this testing
 - Slot#5 = Front Panel card on extender card for getting the Front Panel and switches above the Chassis frame
 - Slot#6 = CPU Card
 - Slot#7 = Micropolis Floppy Card
 - Slot#8 = QT System Real Time Clock Calendar and AAA NmH batteries
 - Slot#9 = Multi Port Serial/Parallel/Timer (has port set up as Centronics to printer)
 - Slot#10 = open (I used this gap for folding ribbon cables going to the back chassis panel)
- 2) See Proposed Configuration page for my specific PCS80/30 set up

VALIDATION PLANS and RESULTS (In Progress):

These procedures are for me, the PCB designer, to validate and verify.
These are FYI for the general user to know the actions I did to check this PCB design (and to be a note to myself to help me know what I did in the future).

- 1) (PASS) KICAD 10.0.4 Design Rule Checking (DRC) tool
 - ** Drc report for CPU-85_r2c.kicad_pcb **
 - ** Created on 2026-07-06T12:05:31 **
 - ** Report includes: Errors, Warnings **
 - ** Found 0 DRC violations **
 - ** Found 0 unconnected pads **
 - ** Found 0 Footprint errors **
- 2) (PASS) new card must insert and egress with reasonable force from 1 hand into and out of S-100 female receiving connector on S-100 back plane.
- 3) (PASS) new card must slide freely between existing IMSAI S-100 back plane card cage guides.
- 4) (PASS) take 2 blank or production fresh panels, install each into 2 different S-100 female connector manufacturer brands 5 times, visually inspect the female connector's witness marks on the edge connectors for consistent location across all edge pins (I don't want to wear out my IMSAI back plane's connectors, IEEE spec has a 50 insertion cycle minimum)
 - black original IMSAI back plane connector:
 - green Mullen Extended Board:
- 5) (DONE) solder on all parts & sockets after receiving new panels, and install all ICs per Assembly Instructions
- 6) Install different Y2 oscillators to find the upper speed limitation of my PCS80/30. Install EPROM with test program at E000H. Look for abnormal text on video display (this operation might take 1 minute per run, enough time for parts to warm up and shift performance if there's a timing issue).

	PCS80/30	With_ReProFP	IMSAI-8080	ALTAIR8800
Verify Power On Jump (Address goes to DIP target + LED indicator blinks)	FAIL			
2.00 MHz for CPU speed = 1MHz = (Y2 XTAL/2)	x			
8.00 MHz	x			
10.00 MHz	x			
11.00 MHz	x			
12.00 MHz	x			
14.00 MHz	x			
15.00 MHz	x			
16.00 MHz	x			
PIC12C675 toggling 80C85B pin1 input at 0.5Hz	PASS			

- 7) Keep the computer and CPU card powered for > 90 minutes with Y2 = 16MHz, and take a thermal image(s) to determine if any components are above more than 25 degrees above ambient (higher temperatures indicate abnormal or unexpected loading)
- 8) change jumper from HOLD Type1 to HOLD Type2, and observe if there is an observable difference in operation
- 9) Plug in Don Caprio's repro version of IMSAI Front Panel and observe what features operate and which do not operate. These tests are done with Y2 = 2MHz.

	PCS80/30	PCS80/30	IMSAI-8080	ALTAIR8800
	Without_ReProFP	With_ReProFP		
RESET				
EXT.RESET (this only effects IO cards not CPU card)	X			
STOP	x			
RUN	x			
SINGLE STEP	x			
EXAMINE	x			
EXAMINE NEXT	x			
LOAD	x			
LOAD NEXT	x			

- 10) ask S100 Vintage Computer Group member with an Altair to try out and return test panel since I don't have a MITS/Altair8800.

VALIDATION PLAN & RESULTS

Sheet: /VALIDATION PLANS/
File: Validation_sch.kicad_sch

Title:

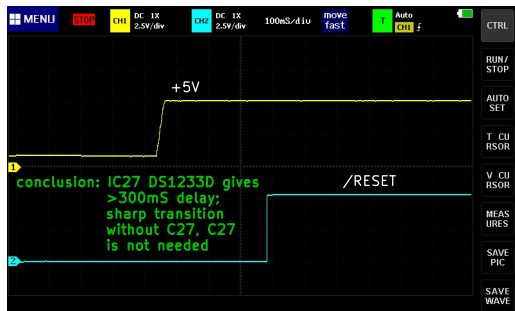
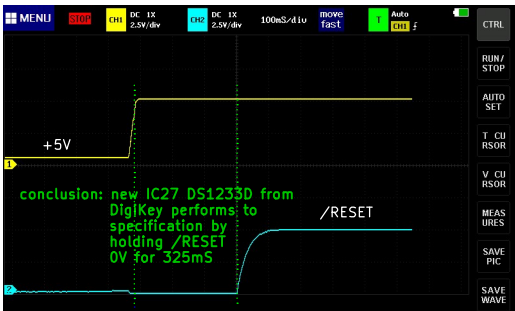
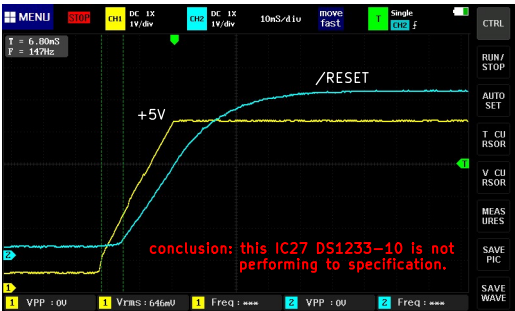
Size: A4	Date: 29 September 2026, greglori_mi@wowway.com	Rev: rev 1.05
KiCad E.D.A. 10.0.4		Id: 13/17

VALIDATION: RESET, SYSTEM CLOCK & PSYNC

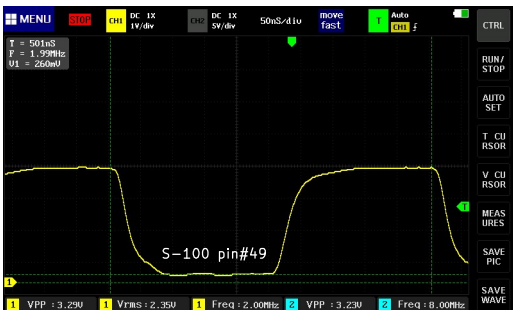
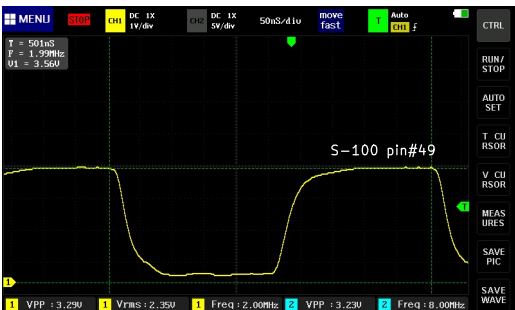
This page was created because when I first powred up the sample CPU85 panel Rewv 1.04.2

DEBUGGING and OBSERVATIONS:

- 11) (FIXED) Found 3 folded under PDIP pins on logic ICs... this happens even to me.
12) (FAIL) (FIXED) CPU stuck in RESET... IC12-36 stuck at Logic 0. IC12-36 resistance to VR1-2 Ground = 9 Ohms.
removed C27, nope IC12-36 to Ground still 9 Ohms. Removed IC27, OK IC12-36 to Ground is 10K Ohms. New DS1233-10 installed, but 350mS delay from DS1233 is not happening.
Only 6.8mS Vcc power up delay not 350mS & no Vcc=4.5V threshold to RESET release is occurring. New DS1233D from DigiKey (9/5/2026) works to specification, desired >300mS /RESET delay.



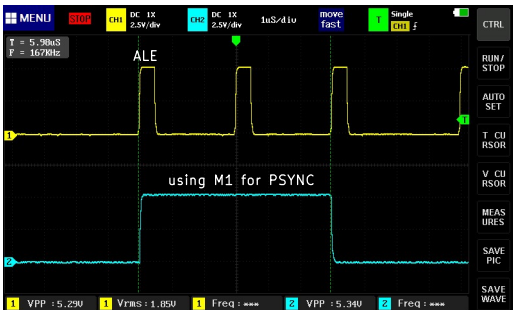
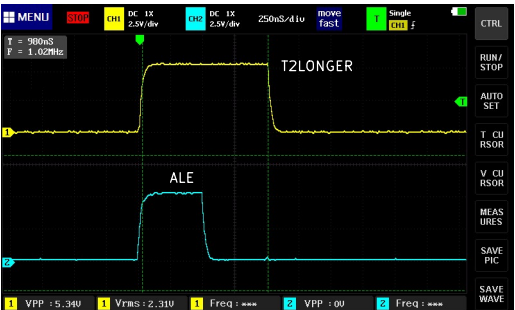
- 13) (PASS) 2.00MHz output at S-100 pin 49 is OK with 500nS period, but oscilloscope is not fast enough for recording the edges very well. This signal is for I/O ports like 8251 Clk_In and some floppy cards.



Yellow trace left is Vhi = 3.56V, good strong logic 1
Yellow trace right is Vlo = 0.25, good strong logic 0

conclusion: 74HCT367 is providing this signal correctly.
changing from 100MHz scope probes to 250MHz probes didn't change recorded wave form.

- 14) (PASS) PSYNC source waveforms, M1, ALE and extended ALE; ALE is normal with 2MHz XTAL input; T2LONGER is expected 2xALE duration; but I did not expect M1 to last while collecting immediate oprand byte(s).



MEASUREMENTS & CAPTURES RESET, SYSTEM CLOCK AND PSYNC

Sheet: /VALIDATION: RESET, SYS CLOCK AND PSYNC/
File: Captures_sch.kicad_sch

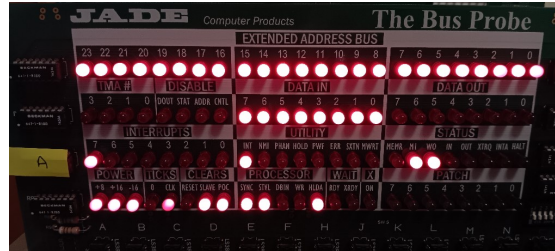
Title:

Size: A4 Date: 17 September 2026, greglori_mi@wowway.com Rev: rev 1.05
KiCad E.D.A. 10.0.4 Id: 14/17

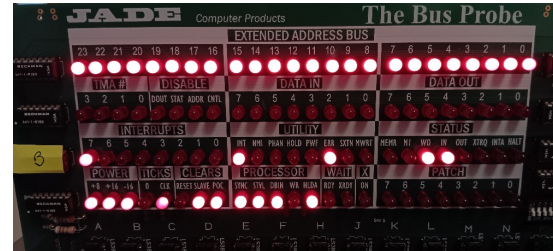
VALIDATION: U24 CPU STATE DECODING

DEBUGGING and OBSERVATIONS:

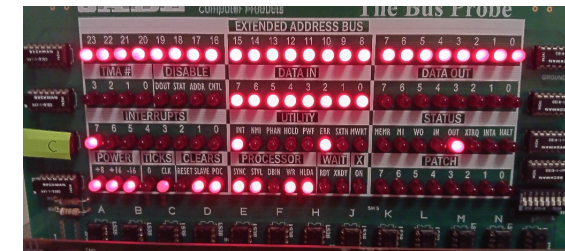
15) (PASS) Since my PCS80/30 does not have multiple CPUs or card(s) using the DMA function, there is no activity on HOLD_TYPE1 and HOLD_TYPE2 signals, thus no difference in selection performance in my system.



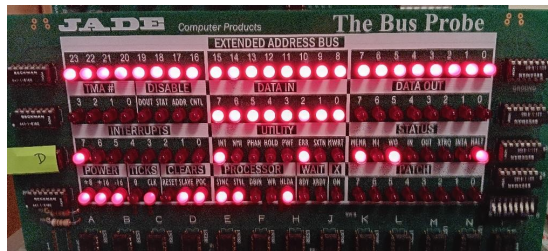
16) (PASS) BUS IDLE (RST or TRAP) State with sm1, U23 Rev3



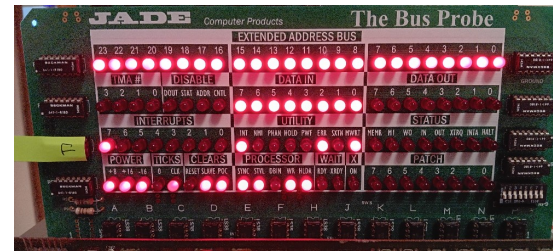
17) (PASS) sINP (Input Read) State, U23 Rev3



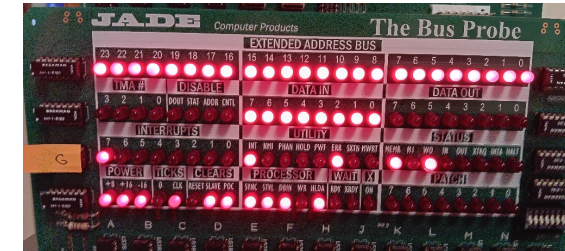
18) (PASS) sOUTP (Output Write) State with sWO OFF, U23 Rev3



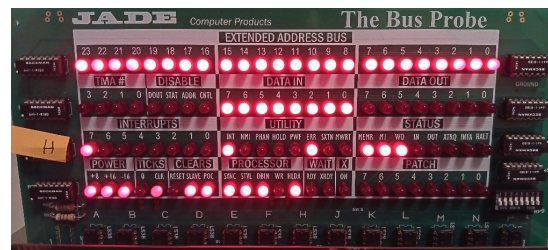
19) (PASS) sHLTA (HALT) State with MEMR, U23 Rev3



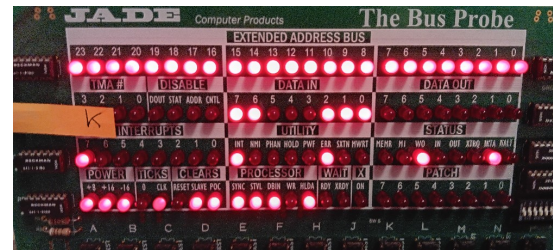
20) (PASS) sMWRT (Memory Write) State with sWO OFF, U23 Rev3



21) (PASS) sMEMR (Memory Read) State, U23 Rev3



22) (PASS) sm1 (Op Code Fetch) State with sMEMR, U23 Rev3



23) (PASS) INT ACK (INA) State when /INTA=0, U23 Rev3

STATUS WORD CHART		TYPE OF MACHINE CYCLE									
		DATA READ	STATUS	INSTRUCTION FETCH	MEMORY READ	MEMORY WRITE	STACK READ	STACK WRITE	INTERRUPT	INTERRUPT ACKNOWLEDGE	INTERRUPT CLEAR
		1	2	3	4	5	6	7	8	9	10
D0	INTA	0	0	0	0	0	0	0	0	0	0
D1	WO	1	1	0	0	0	0	0	0	0	0
D2	STACK	0	0	0	0	0	0	0	0	0	0
D3	HLTA	0	0	0	0	0	0	0	0	0	0
D4	OUT	0	0	0	0	0	0	0	0	0	0
D5	M1	0	0	0	0	0	0	0	0	0	0
D6	INP	0	0	0	0	0	0	0	0	0	0
D7	MEMR	0	0	0	0	0	0	0	0	0	0

Table 2-1. 8080 Status Bit Definitions

Notes:

Table 2-1 is copied from Intel MCS80 CPU and Peripherals Data Sheets Book 1975 page 10. The 8085 lacks STACK READ and STACK WRITE flags (not used by any products so they these were no longer supported).

This validation was done with IC12 removed, and using DIP switches to toggle the CPU RD, WR, HLDA, IOM, etc signals directly.

- = PASS or desired logic state LED illuminated on Jade Bus Probe board (U24 decoding)
- = PASS or desired other logic state LED OFF on Jade Bus Probe board (not highlighted)
- = FAIL or not desired logic state (U24 decoding failures rev1 and rev2 had some)
- S-100 systems did not use CPU State#10, HALT with wake up from STOP mode.
- x = DON'T CARE or logic state not actually used by S-100 cards

MEASUREMENTS & CAPTURES U24 CPU State Verification

Sheet: /VALIDATION: U24 CPU STATE DECODING/
File: More_Captures_sch.kicad_sch

Title:

Size: A4 Date: 17 September 2026, greglori_mi@wowway.com

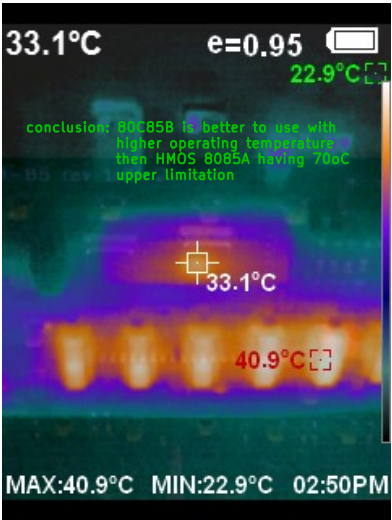
KiCad E.D.A. 10.0.4

Rev: rev 1.05

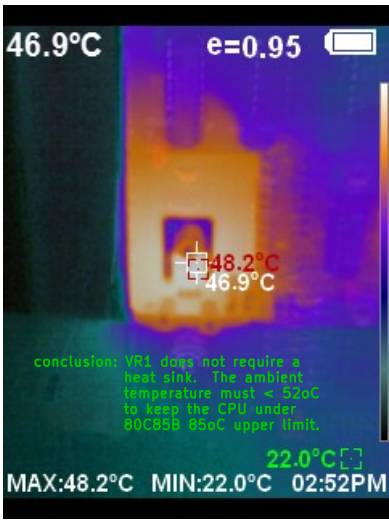
Id: 15/17

VALIDATION: THERMAL IMAGES

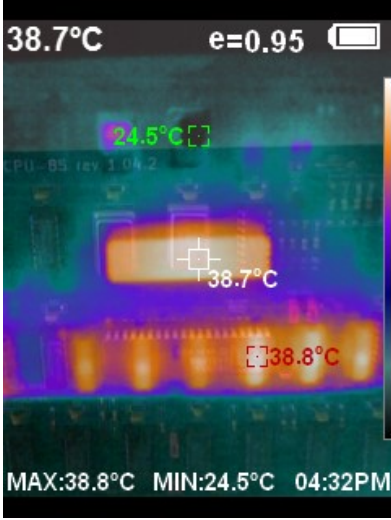
DEBUGGING and OBSERVATIONS:
24) (PASS) The thermal images are to verify no parts are over heating or being over stressed >30 (was 25) degrees above ambient. Y2 = 16MHz for maximum toggling using 80C85B. Y2 = 12MHz for maximum toggling using 8085A-1.



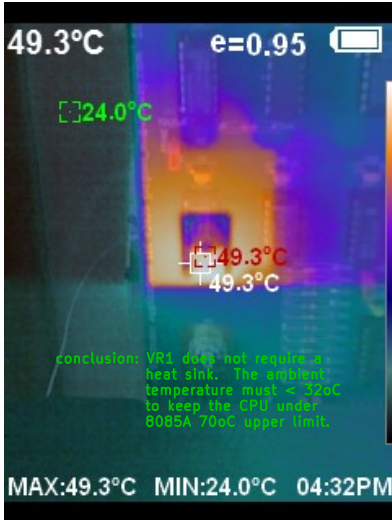
25) (PASS) IC12 CPU 80C85B @ 8MHz 33.1oC IC16 thru IC20 are warm, Data and Address Bus activity is high, IC20 is 40.9oC, all < 47.9oC



26) (PASS) VR1 regulator is 48.2oC at test limit the T0220 case is warm to touch, but not an injury risk, +30 wouold be better open air fail threshold; notice heat spreading through Ground planes but trapped by PCB masking ... expose more bare metal Ground Plane around VR1; board rev 1.04.2



27) (PASS) IC12 CPU 8085A-1 @ 6MHz 38.7oC IC18 thru IC20 are warm, Data and Address Bus activity is moderate, IC18 is 38.8oC, all < 49.5oC



28) (PASS) VR1 regulator is 49.3oC at test limit the T0220 case is warm to touch, but not an injury risk, +30 wouold be better open air fail threshold; notice heat spreading through Ground planes but trapped by PCB masking ... again expose more bare metal Ground Plane around VR1; board rev 1.04.2

this space reserved for thermal image of VR1 of 1.05 with more exposed Ground plane

MEASUREMENTS & CAPTURES THERMAL IMAGES

Sheet: /VALIDATION: THERMAL IMAGES/ File: More_Captures3_sch.kicad_sch		
Title:		
Size: A4	Date: 29 September 2026, greglori_mi@wowway.com	Rev: rev 1.05
KiCad E.D.A. 10.0.4		Id: 16/17

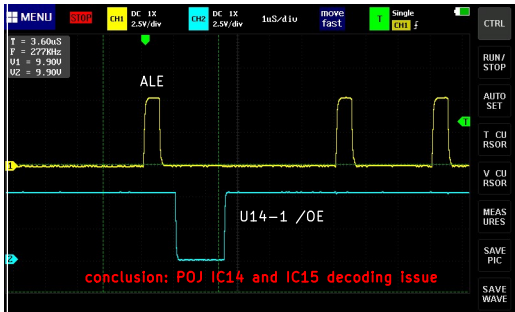
VALIDATION: Dout, ADDRESS, CPU STATE, Din BUS ENABLES

DEBUGGING and OBSERVATIONS:

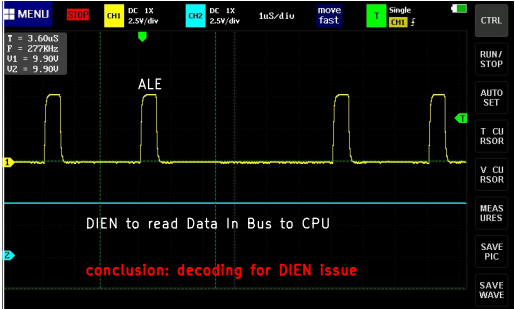
31) These checks are to verify that no 2 bus drivers are enabled at the same time to when driving Dout and Din into the CPU.

Expectations:

- * CPU State to S-100 Dout only when PSYNC=1, CPU Data Lines to Dout when PSYNC=0
- * Din to CPU when DIEN=1 & POJ=0, when POJ=1, Din must be disabled, allowing U14 and U15 to drive into the CPU as the data sources
- * Address 0-15 always output to S-100 Bus (this test system does not have any other CPUs or DMA devices to drive the address bus)
- * with POJ enabled, POJ LED illuminates briefly, verify first instruction is JMP E000 or C3,00,E0 bytes from U14 and U15, fourth byte is from E000.
- * POJ LED never illuminates when disabled when SW1-1 = OFF.



32) (FAIL) Expected 2 U14 /OE on first 2 ALE pulses;
First /OE pulse is OK, but missing the second



33) (FAIL) Expected read pulsing on /DIEN, none ever happen

PRELIMINARY: THIS PAGE IS UNDER CONSTRUCTION,
NEW SCOPE CAPTURES TO BE COLLECTED AFTER PCB REVISION.

MEASUREMENTS & CAPTURES S-100 Bus Enables

Sheet: /VALIDATION: BUS DRIVER ENABLES/
File: More_Captures2_sch.kicad_sch

Title:

Size: A4

Date: 6 October 2026, greglori_mi@wowway.com

Rev: rev 1.05

KiCad E.D.A. 10.0.4

Id: 17/17